

ContractIQ: Contract Obligation Tracking & Compliance Management Platform

Sprint 3 Deliverable – Database Design & Schema Architecture Document

1. Executive Summary & Objective

The primary objective of **Sprint 3 – Database Design** is to define a robust, scalable, and normalized relational database structure for **ContractIQ**, a full-stack Contract Obligation Tracking & Compliance Management Platform.

This design serves as the foundation for:

- Managing complex legal contract lifecycles (Draft → Review → Approval → Active → Expired/Terminated).
- Tracking obligations and compliance levels with granular status monitoring.
- Automating renewal monitoring, expiry alerts, and notification scheduling.
- Maintaining strict regulatory compliance through immutable audit logging and version control.
- Powering analytics dashboards and export reports.

Target RDBMS: **PostgreSQL 15+**

Backend ORM Mapping: **SQLAlchemy 2.0+ (Python / FastAPI)**

2. Table Specifications & Column Definitions

The application database consists of **9 core tables**:

2.1 Table: users

Purpose: Manages user authentication, profile data, department assignments, and Role-Based Access Control (RBAC).

Column Name	Data Type	Constraints	Key Type	Description
user_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique identifier for each user
first_name	VARCHAR(100)	NOT NULL	-	User's first name
last_name	VARCHAR(100)	NOT NULL	-	User's last name
email	VARCHAR(255)	NOT NULL, UNIQUE	-	Work email used for login & notifications
hashed_password	VARCHAR(255)	NOT NULL	-	Bcrypt password hash
role	VARCHAR(50)	NOT NULL	-	Role (Administrator, Legal Manager, Compliance Officer, Contract Manager, Department Head, Employee)
department	VARCHAR(100)	NULLABLE	-	User's department (e.g., Legal, Procurement, HR, Finance)

Column Name	Data Type	Constraints	Key Type	Description
phone_number	VARCHAR(20)	NULLABLE	-	Contact phone number for SMS notifications
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	-	Account active status flag
created_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Account creation timestamp
updated_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Account update timestamp

2.2 Table: contracts

Purpose: Stores core contract metadata, category, lifecycle state, counterparty information, financial value, and key milestone dates.

Column Name	Data Type	Constraints	Key Type	Description
contract_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique identifier for the contract
contract_number	VARCHAR(50)	NOT NULL, UNIQUE	-	Human-readable unique contract reference number
title	VARCHAR(255)	NOT NULL	-	Title/Name of the contract
category	VARCHAR(50)	NOT NULL	-	Category (Employment Contracts, Vendor Contracts, Service Agreements, Lease Agreements, Purchase Agreements, Partnership Agreements, Confidentiality Agreements)
status	VARCHAR(50)	NOT NULL, DEFAULT 'Draft'	-	Status (Draft, Under Review, Approved, Active, Expired, Terminated)
counterparty_name	VARCHAR(255)	NOT NULL	-	External vendor/party name
counterparty_email	VARCHAR(255)	NULLABLE	-	Contact email for the external party
effective_date	DATE	NOT NULL	-	Contract start date
expiration_date	DATE	NOT NULL	-	Contract termination/expiry date
total_value	NUMERIC(15,2)	NOT NULL, DEFAULT 0.00	-	Financial value of the contract
currency	VARCHAR(3)	NOT NULL, DEFAULT 'USD'	-	ISO Currency code (e.g. USD, EUR, INR)

Column Name	Data Type	Constraints	Key Type	Description
created_by	UUID	NOT NULL	FK (users)	User who created the contract entry
assigned_to	UUID	NULLABLE	FK (users)	Primary owner responsible for managing contract
approved_by	UUID	NULLABLE	FK (users)	Legal/Admin user who approved the contract
created_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Record creation timestamp
updated_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Record update timestamp

2.3 Table: contract_versions

Purpose: Supports version control and document repository storage paths (S3/Cloud storage) for original documents and updated amendments.

Column Name	Data Type	Constraints	Key Type	Description
version_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique version record identifier
contract_id	UUID	NOT NULL	FK (contracts)	Reference to parent contract
version_number	VARCHAR(20)	NOT NULL	-	Version tag (e.g. v1.0, v1.1, v2.0)
file_name	VARCHAR(255)	NOT NULL	-	Original uploaded file name
file_path	TEXT	NOT NULL	-	S3/Cloud Storage object key/URL
file_size_bytes	BIGINT	NOT NULL	-	Size of file in bytes
file_hash	VARCHAR(64)	NOT NULL	-	SHA-256 integrity hash
change_summary	TEXT	NULLABLE	-	Release notes / summary of changes
uploaded_by	UUID	NOT NULL	FK (users)	User who uploaded this document version
created_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Version upload timestamp

2.4 Table: obligations

Purpose: Tracks granular legal/business obligations, milestone due dates, assigned owners, obligation types, and compliance levels.

Column Name	Data Type	Constraints	Key Type	Description
obligation_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique obligation identifier
contract_id	UUID	NOT NULL	FK (contracts)	Associated contract
title	VARCHAR(255)	NOT NULL	-	Brief title of obligation
description	TEXT	NULLABLE	-	Detailed description of requirement
obligation_type	VARCHAR(50)	NOT NULL	-	Type (Payment Obligations, Delivery Commitments, Reporting Requirements, Renewal Conditions, Service Level Agreements, Legal Compliance Requirements)
compliance_level	VARCHAR(50)	NOT NULL, DEFAULT 'Pending'	-	Current compliance status (Compliant, Pending, Delayed, Non-Compliant, High Risk)
due_date	DATE	NOT NULL	-	Deadline date for obligation fulfillment
completed_at	TIMESTAMPZ	NULLABLE	-	Timestamp when obligation was fulfilled
assigned_to	UUID	NOT NULL	FK (users)	Person responsible for fulfilling obligation
penalty_description	TEXT	NULLABLE	-	Contractual financial or legal penalty for breach
created_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Record creation timestamp
updated_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Record update timestamp

2.5 Table: renewals

Purpose: Manages contract renewal schedules, notice period windows, auto-renewal preferences, and renewal workflow progression.

Column Name	Data Type	Constraints	Key Type	Description
renewal_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique renewal entry identifier
contract_id	UUID	NOT NULL	FK (contracts)	Parent contract subject to renewal
renewal_notice_date	DATE	NOT NULL	-	Date by which notice must be given
renewal_date	DATE	NOT NULL	-	Official renewal date
renewal_status	VARCHAR(50)	NOT NULL, DEFAULT 'Upcoming'	-	Status (Upcoming, In Progress, Renewed, Expired, Cancelled)
auto_renew	BOOLEAN	NOT NULL, DEFAULT FALSE	-	Whether contract automatically renews
notice_period_days	INTEGER	NOT NULL, DEFAULT 30	-	Days required for advance termination notice
new_expiration_date	DATE	NULLABLE	-	Updated contract expiration date post-renewal
processed_by	UUID	NULLABLE	FK (users)	User handling renewal negotiation/approval
notes	TEXT	NULLABLE	-	Renewal terms or negotiation notes
created_at	TIMESTAMP PTZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Timestamp of creation
updated_at	TIMESTAMP PTZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Timestamp of update

2.6 Table: notifications

Purpose: Stores automated system alerts, obligation due notifications, renewal reminders, and multi-channel delivery states (Email, SMS, In-App).

Column Name	Data Type	Constraints	Key Type	Description
notification_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique notification identifier
user_id	UUID	NOT NULL	FK (users)	Recipient user
contract_id	UUID	NULLABLE	FK (contracts)	Related contract (if applicable)

Column Name	Data Type	Constraints	Key Type	Description
obligation_id	UUID	NULLABLE	FK (obligations)	Related obligation (if applicable)
renewal_id	UUID	NULLABLE	FK (renewals)	Related renewal event (if applicable)
title	VARCHAR(255)	NOT NULL	-	Short alert header
message	TEXT	NOT NULL	-	Body text of the notification
channel	VARCHAR(50)	NOT NULL, DEFAULT 'In-App'	-	Channel (In-App, Email, SMS)
notification_type	VARCHAR(50)	NOT NULL	-	Category (Renewal Reminder, Obligation Due, Compliance Alert, Contract Approval)
is_read	BOOLEAN	NOT NULL, DEFAULT FALSE	-	User read state for in-app inbox
sent_at	TIMESTAMPTZ	NULLABLE	-	Actual delivery timestamp
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Notification generation timestamp

2.7 Table: reports

Purpose: Persists generated compliance summaries, renewal audits, obligation metrics, and file export parameters (PDF/Excel).

Column Name	Data Type	Constraints	Key Type	Description
report_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique report metadata identifier
report_name	VARCHAR(255)	NOT NULL	-	Display title of the report
report_type	VARCHAR(50)	NOT NULL	-	Report focus area (Contract Report, Compliance Report, Renewal Report, Obligation Report, Audit Report)
export_format	VARCHAR(20)	NOT NULL	-	Format (PDF, Excel, CSV)
filter_parameters	JSONB	NULLABLE	-	Filters used (date ranges, category, status)
file_url	TEXT	NOT NULL	-	Download path/S3 link to generated document
generated_by	UUID	NOT NULL	FK (users)	User who requested the report
generated_at	TIMESTAMPTZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Report generation timestamp

2.8 Table: audit_logs

Purpose: Provides security-grade, append-only audit tracking for data mutations, security access events, and approval actions.

Column Name	Data Type	Constraints	Key Type	Description
audit_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique audit record identifier
user_id	UUID	NULLABLE	FK (users)	User performing the action (NULL for automated system jobs)
action	VARCHAR(100)	NOT NULL	-	Action name (e.g. CONTRACT_CREATED, STATUS_UPDATED, DOCUMENT_DOWNLOADED, OBLIGATION_FULFILLED)
entity_type	VARCHAR(50)	NOT NULL	-	Affected table entity (e.g. contracts, obligations, renewals)
entity_id	UUID	NOT NULL	-	Primary Key ID of the targeted record
old_values	JSONB	NULLABLE	-	Snapshot of record state prior to edit
new_values	JSONB	NULLABLE	-	Snapshot of record state post-edit
ip_address	VARCHAR(45)	NULLABLE	-	Client IPv4 / IPv6 address
user_agent	TEXT	NULLABLE	-	Browser / API client identification string
timestamp	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Precise event timestamp

2.9 Table: activities

Purpose: Powers real-time dashboard feeds, user action streams, and collaboration comments attached to contracts.

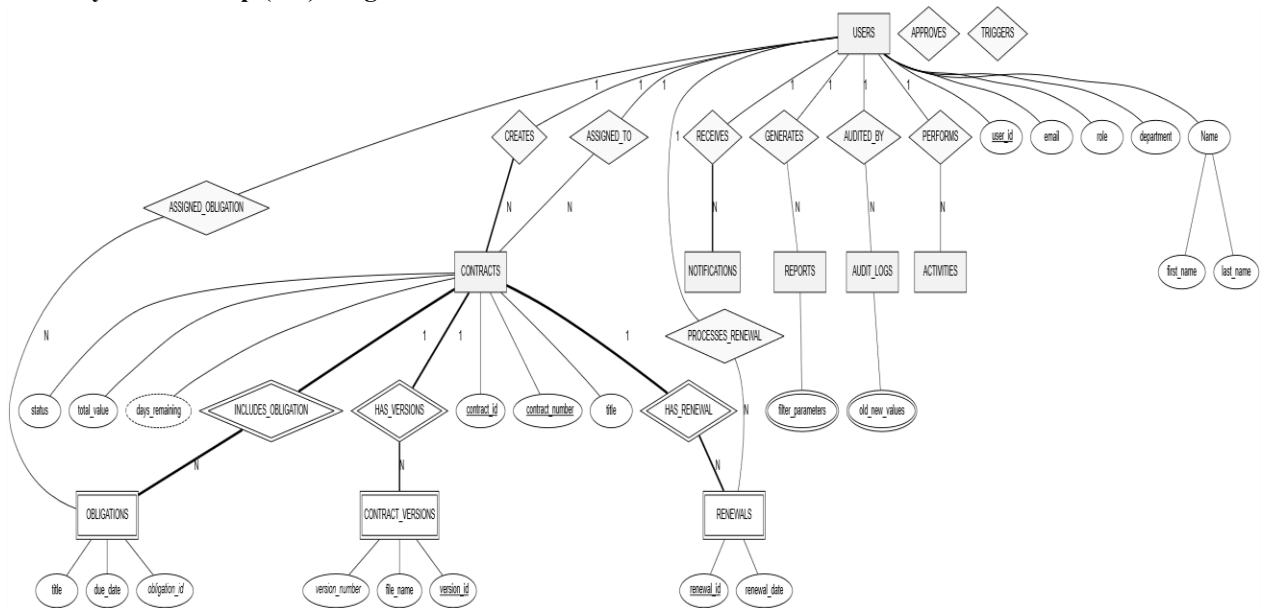
Column Name	Data Type	Constraints	Key Type	Description
activity_id	UUID	NOT NULL, DEFAULT gen_random_uuid()	PK	Unique activity log ID
user_id	UUID	NOT NULL	FK (users)	User triggering activity
contract_id	UUID	NULLABLE	FK (contracts)	Targeted contract (if applicable)
activity_type	VARCHAR(50)	NOT NULL	-	Event type (COMMENT_ADDED, DOCUMENT_VIEWED, STATUS_CHANGED, RENEWAL_REQUESTED)

Column Name	Data Type	Constraints	Key Type	Description
description	TEXT	NOT NULL	-	Human-readable activity message
created_at	TIMESTAMPZ	NOT NULL, DEFAULT CURRENT_TIMESTAMP	-	Activity creation timestamp

3. Entity Relationships Summary

Parent Table	Child Table	Relationship Type	Foreign Key Column	Description & Multiplicity
users	contracts	One-to-Many (1:N)	created_by, assigned_to, approved_by	One user can create, manage, or approve multiple contracts.
contracts	contract_versions	One-to-Many (1:N)	contract_id	One contract can have multiple document versions over time.
contracts	obligations	One-to-Many (1:N)	contract_id	One contract contains multiple compliance obligations.
users	obligations	One-to-Many (1:N)	assigned_to	One user can be assigned responsibility for multiple obligations.
contracts	renewals	One-to-Many (1:N)	contract_id	One contract can undergo multiple renewal cycles across its lifetime.
users	renewals	One-to-Many (1:N)	processed_by	One manager can process multiple contract renewals.
users	notifications	One-to-Many (1:N)	user_id	One user receives multiple automated notifications.
contracts	notifications	One-to-Many (1:N)	contract_id	A contract can trigger multiple reminders and alerts.
obligations	notifications	One-to-Many (1:N)	obligation_id	An obligation can generate multiple due/breach alerts.
renewals	notifications	One-to-Many (1:N)	renewal_id	A renewal schedule can trigger multiple automated reminders.
users	reports	One-to-Many (1:N)	generated_by	One user can request/generate multiple report exports.
users	audit_logs	One-to-Many (1:N)	user_id	One user performs many auditable actions throughout the system.
users	activities	One-to-Many (1:N)	user_id	One user produces many activity feed events.
contracts	activities	One-to-Many (1:N)	contract_id	A contract accumulates an interactive activity history stream.

4. Entity Relationship (ER) Diagram



5. Architectural Explanation: How the Tables Work Together

The **ContractIQ** database schema is engineered around a centralized hub model where the contracts table acts as the primary focal entity, interconnecting governance, compliance monitoring, document repository management, and automated notifications. When a user (authenticated via users with appropriate RBAC privileges such as *Legal Manager* or *Contract Manager*) initializes a new contract, a record is inserted into contracts, and an initial document file payload creates a record in contract_versions with file path pointers and integrity hashes.

As the contract moves into an active status, granular compliance terms are registered under obligations, linking each requirement (e.g., SLA, payment, delivery) back to the parent contracts record while assigning accountability to specific team members via users.user_id. Concurrently, contract lifecycle milestones (such as advance termination windows and auto-renewal dates) are tracked in renewals. Automated background workers poll obligations and renewals against deadline windows to populate notifications (supporting In-App, Email, and SMS delivery via Twilio/SMTP).

For organizational visibility and compliance auditing, all write operations across contracts, obligations, and renewals automatically record immutable change snapshots in audit_logs (tracking who changed what, when, and from what IP address), while real-time user interactions drive the activities feed on dashboard screens. Finally, aggregated analytical reports requested by Legal Officers or Department Heads read across these entities to synthesize metrics into reports files (available for PDF/Excel download).

